U.S. DEPARTMENT OF AGRICULTURE
WASHINGTON, DC 20250

DEPARTMENTAL REGULATION	NUMBER: DR 4400-007
SUBJECT: USDA Biorisk Management Policy	DATE: September 3, 2020
OPI: Animal and Plant Health Inspection Service (APHIS)	EXPIRATION DATE: September 3, 2025

1. PURPOSE

This Departmental Regulation (DR) establishes the policy of the United States Department of Agriculture (USDA) policy, requirements, and responsibilities for administering a comprehensive biorisk management program across the Department.

2. SCOPE

a. This DR applies to all USDA Mission Areas, agencies, staff offices, employees, appointees, contractors, and others who work for, or on behalf of, USDA.

b. This DR provides guidance to USDA Mission Areas, agencies, and staff offices conducting operations that require employees, contractors, cooperators, students, or visitors who may have potential exposure to known or potentially hazardous biological agents or toxins. Examples of such operations include, but are not limited to:

(1) Research and diagnostic laboratory operations, animal holding facilities (including insectaries and aquatic species holding areas), plant propagation facilities (e.g., greenhouses, headhouses, and quarantine facilities), and field research settings where hazardous biological agents or toxins are collected, propagated, manipulated, stored, or studied.

(2) Environmental or other laboratory operations that receive samples suspected of containing hazardous biological agents or toxins (biohazardous agents) to include High Consequence Biological Agents (HCBA) or Biological Select Agents and Toxins (BSAT). For example, a soil analysis laboratory where samples may contain Bacillus anthracis spores or mycotoxin-producing fungi; or a chemical residue analysis laboratory that tests animal sera and tissues, or biological and microbial products (e.g., vaccines).

3. SPECIAL INSTRUCTIONS/CANCELLATIONS

a. This regulation supersedes DR 4400-007, *Biological Safety Program*, dated May 19, 2006.

b. This policy is effective immediately and remains in effect until superseded or expiration.

c. All Mission Areas, agencies, and staff offices will align their policies and procedures with this DR within 6 months of the publication date.

d. The USDA Biorisk program management procedures and coordination processes will be provided in the forthcoming Departmental Manual (DM) 4400-007, *USDA Biorisk Management Manual*.

4. POLICY

Life sciences research, disease diagnostic development, food safety, and the improvement of our crops and livestock serve to actively foster the long-term health, security, wellness of the public, animals (domestic and wildlife), plants, the environment, and our economy. The safe and responsible execution of life sciences research and its ancillary disciplines have been indispensable in maintaining and improving our Nation's capabilities to identify and mitigate the risks of infectious diseases and environmental risks, whether naturally occurring, accidental, or intentional.

a. It is USDA policy that each Mission Area, agency, and staff office that fits within the scope of this policy develop and implement a comprehensive biorisk management policy that will:

(1) Require agencies to establish a process to assess and mitigate risks to an acceptable level for research and diagnostic activities related to work with biological agents or toxins, as explained later in this policy;

(2) Minimize occupational exposures to infectious materials including zoonotic, opportunistic, or otherwise hazardous biological agents or toxins. These exposures could originate from food, research or commercial agricultural animals, wildlife, specimens collected from those sources or environmental sources, or waste materials from those activities;

(3) Minimize occupationally-acquired infections;

(4) Minimize the release of infectious materials into the environment, movement or cross-contamination of infectious biological agents or toxins within and between laboratory and field research locations, production locations visited by USDA employees, and agriculture commodities/products or plant nursery stocks;

(5) Ensure appropriate levels of physical protection against unauthorized access, theft, diversion, or loss of custody of pathogens or other high-value research assets;

(6) Ensure adequate resources (e.g., staff, funding, training, equipment, and tools) are dedicated to the development, management, operation, and evaluation of a comprehensive biorisk management program at all levels, and

(7) Ensure Mission Areas, agencies, and staff offices biorisk management programs are reviewed as defined by individual agencies and that the criteria for their review are based on established guidelines (e.g., the Centers for Disease Control and Prevention (CDC) *Biosafety in Microbiological and Biomedical Laboratories*, current edition (BMBL)) or other national or international biorisk management related consensus standards (some consensus standards may include, but are not limited to, ISO/ANSI (International Organization for Standardization/American National Standards Institute), OIE (World Organization for Animal Health), WHO (World Health Organization), and NSF (National Sanitation Foundation)), as appropriate, or scientifically sound criteria in addition to specific agency goals, objectives, or criteria. The results of the reviews are documented and retained to drive appropriate changes or adjustments to the relevant program with the goal of continual improvement.

b. To ensure that Departmental leadership is provided with the necessary programmatic reports and analyses to provide sufficient oversight of biorisk management policy across USDA. Agency reports will be submitted at least annually and all significant incidents will be submitted within 2 business days to the appropriate Mission Area Under Secretary and to the USDA Joint Committee on Biorisk Management Policy (JCBMP) Executive Secretary for review by the Operational Biorisk Subcommittee (OBC). Significant incidents include:

(1) Two or more individuals exposed to an infectious or zoonotic agents, or toxins;

(2) Suspected or known occupational acquired infection (laboratory- or field-acquired infection (LAI));

(3) Reportable environmental release or accidental infections of animals;

(4) Significant security breaches (e.g., theft);

(5) Regulatory citations (notice of violations); and

(6) Critical inspection findings (internal or external) that would require immediate Departmental attention and/or action.

c. The subcommittee will use these reports to evaluate trends and lessons learned, share feedback with individual agencies and programs, and facilitate subsequent reviews as required. Summaries of report findings and the results of these evaluations will be shared at least annually with USDA senior leadership via the JCBMP.

5. POLICY EXCEPTIONS

a. All USDA Mission Areas, agencies, and staff offices are required to conform to this policy. In the event that a specific policy requirement cannot be met as explicitly stated, or conformance with a specific policy requirement will adversely affect an agency's or staff office's ability to fulfill its normal operations or functions, agencies and staff offices must submit a waiver request. The waiver request will explain the reason for the request, identify compensating, mitigating controls or actions that meet the intent of the policy, and identify how the compensating, mitigating controls or actions provide a similar or greater level of safety and security or compliance than the policy requirement.

b. Requests for waivers will be considered on a case-by-case basis and tracked by each agency. In the majority of cases, waiver requests can be resolved between the field and staff offices. Waiver requests must be submitted in the form of a decision memorandum. Agencies and staff offices should address all policy waiver request memorandums to their individual biosafety, physical security, incident response, personnel security, and/or cybersecurity Point of Contact (POC), as appropriate. Each agency will provide routine updates to the JCBMP's OBC.

c. Unless otherwise specified, agencies (e.g., Agency Biosafety Officer or designee) and staff offices shall review and renew approved policy waivers every fiscal year. Approved waivers shall be tracked by each agency as a plan of action and milestones (POA&M) item if the mitigation proposed was working toward policy compliance over time. Approved waivers for specialized equipment and/or facilities requirements or enhancements with a continuing need will simply be tracked on an approved list.

d. The appropriate agency-level office (e.g., Agency Biosafety Officer or designee) shall monitor and approve waivers to this policy internally. Consultation with additional Departmental resources or subject matter experts on changing conditions as waivers are monitored will be pursued on a case-by-case basis.

e. The written waiver request decision memorandum will include:

(1) Request for Exception, addressed to the appropriate agency resource;

(2) Name of submitting agency or staff office;

(3) Name and contact information of submitting person; and

(4) Description of the policy exception request:

(a) Justification to show good cause for the exception. The request should document the justifications for the exception; and

(b) The impact of granting versus not granting the request.

6. ROLES AND RESPONSIBILITIES

a. The Secretary of Agriculture will:

(1) Ensure processes are in place for the development, publication, and active maintenance of Departmental policies, regulations, and general requirements for biorisk management;

(2) Provide management and oversight activities to ensure that agencies are in compliance with this DR in implementing its biorisk management program; and

(3) Ensure that indicators, beliefs, and values developed in USDA policies and regulations assert USDA ownership of its biorisk management activities. The USDA JCBMP and its OBS were established to provide resources to the Department.

b. APHIS Administrator will:

(1) Be responsible for the management of the APHIS Biorisk Management Program, or designee; and

(2) Review and update this Departmental Regulation, or designee, in collaboration with the JCBMP.

c. The USDA JCBMP Co-Chairs will:

(1) Be co-chaired by the Office of Chief Scientist (OCS) Director and Office of Homeland Security (OHS) Director or their designees;

(2) Be governed by the provisions of the JCBMP Charter regarding membership, operations, and administration;

(3) Provide compliance oversight and advice to agency's biorisk management or safety staff on biorisk management initiatives directed or recommended by the White House, the Government Accountability Office (GAO) , the Federal Select Agent Program (FSAP) which is co-managed by the CDC and USDA, and other expert advisory panels;

(4) Review and provide clearance on USDA inputs related to biorisk management national reports (e.g., Dual Use Research of Concern (DURC)) before they are delivered to the White House, Congress, or interagency partners;

(5) Identify potential trends that may highlight recurring laboratory- or field-safety or security issues, and share that information with laboratory personnel;

(6) Review and routinely report to senior Departmental officials summaries of laboratory inspections (e.g., applicable regulatory agency inspections, and Agency or Departmental internal inspections) and laboratory incidents or accidents as defined in this DR;

(7) Report the status of agency biorisk management programs to Departmental leadership; and

(8) Review and as necessary revise biorisk management program elements in response to actual events, potential trends, inspection findings, or any other significant event or initiative that relates to biological risk, personnel suitability, or information security.

d. The USDA JCBMP OBC Co-Chairs will:

(1) Be Chaired or Co-Chaired by an agency Biosafety Officer(s) or designee;

(2) Be governed by the provisions of the JCBMP OBC Charter regarding membership, operations, and administration;

(3) Serve as a Departmentwide resource for biosafety and biosecurity issues as they relate to laboratory operations, and provide oversight to help ensure USDA compliance with applicable laws, standards, and guidance;

(4) Facilitate assistance to USDA agencies when they develop and update their agency-specific biorisk management policies, to include field-biosafety policies, as

applicable, that are relevant to laboratory operations, field studies and operations, and identify best practices that should be standardized across all agencies;

(5) Report agency biorisk-related incidents and inspections to the JCBMP as directed;

(6) Develop and revise the Departmentwide biorisk management policy as directed by the JCBMP; and

(7) Review and address any questions or inquiries related with this policy that are received by the OPI.

e. USDA Mission Area, Agency Head, and Staff Office Directors will:

(1) Promote a safety culture throughout the organization that emphasizes biosafety, biosecurity, and responsible conduct in the life sciences;

(2) Develop, publish, and actively maintain agency policies, regulations, and compliance requirements for biorisk management;

(3) Provide support and resources necessary for a robust biorisk management structure within their agency, including incorporating policies, requirements, and standards into agency and staff office capital planning and investment control processes;

(4) Ensure that biorisk management expertise is made available at the agency level and at facilities and locations that possess or use biological agents or toxins. When dedicated biorisk management expertise is not feasible at a facility, location, or worksite, ensure that a collateral duty safety officer is proficient in biorisk management; and

(5) Clearly define and delineate Departmental and agency support in biorisk management policy implementation (e.g., site-specific risk assessment, job hazard analysis (JHA) and training, personnel suitability program support).

f. Deputy Administrators, Associate Deputy Administrators, Assistant Administrators, and Deputy Assistant Administrators (according to agency structure) will:

(1) Promote an organizational culture that emphasizes biosafety, biosecurity, and responsible conduct in the life sciences;

(2) Provide direct assistance to agency administrators to ensure that the agency correctly implements and complies with the policies, procedures, and requirements of all biorisk management and safety, health, and environmental programs;

(3) Develop comprehensive and integrated biorisk management policies, procedures and programs; and

(4) Ensure that all managers and supervisors implement the requirements of biorisk management and safety, health, and environmental programs, as applicable.

g. Principal Investigators, Lead Scientists, and Lab Managers and Supervisors will:

(1) Promote a culture that emphasizes biosafety, biosecurity, and responsible conduct in the life sciences;

(2) Provide resources to implement a robust biorisk management program, including but not limited to, training and monitoring of safety and security policies and programs;

(3) Read and become thoroughly familiar with USDA biorisk management Departmental directives; agency-specific biorisk management program policies, procedures, and biosafety manuals; and applicable Federal, State, and local statutes and regulations;

(4) Conduct comprehensive site-specific risk assessments that will inform the application of biosafety, biosecurity, physical security, incident response, personnel suitability, and/or information security policies and procedures as applicable (seek out Departmental and agency expertise on these fields if it is not available at the site);

(5) Ensure that employees receive initial and recurring specialized job training (e.g., environmental, safety, and security) appropriate to the work they are required to perform;

(6) Conduct refresher training on required schedules and whenever job assignments change, operational procedures change, or employees are not following correct procedures;

(7) Provide employees time to participate in USDA-sponsored biorisk management and safety, health, and environmental initiatives, including training, without restraint, interference, coercion, discrimination, or reprisal;

(8) Assist agency leadership and safety and health staff in operating an effective biorisk management program as it relates to their areas of supervision;

(9) Report, investigate, and document all job-related incidents in a timely manner and in accordance with agency-specific policies, as documented in each location- or facility-specific procedures.

(a) Including accidents, injuries, illnesses, and exposures (including suspected exposures) to infectious biological agents (including zoonotic agents) and biological toxins of concern; and

(b) Security-related incidents and environmental releases of biological agents of agricultural concern, in a timely manner, in accordance with agency policy;

The investigation must include follow-up items and reviews of corrective actions to ensure full and complete implementation of those actions;

(10) Brief employees on biorisk management policies and procedures applicable to their specific worksite, and enforce such policies;

(11) Ensure that proper personal protective equipment (PPE) is available free of charge; that employees are trained in its proper use, maintenance, or disposal; and that it is worn whenever necessary;

(12) Encourage and promote employee suggestions on how to improve biorisk management, safety, and health in the workplace, and take appropriate action on those suggestions; and

(13) Develop a biorisk management plan, job hazard analysis, and standard operating procedures (SOP) to implement agency policies for specific locations and facilities. Supervisors should have the option of establishing requirements above the base requirements defined in agency or location policies based on a scientifically valid, site- or procedure-specific risk assessment.

h. Biorisk management staff members (which may include Biosafety Officers, safety and health or scientific staff as an adjunct duty) will:

(1) Promote a culture that emphasizes the importance of biosafety, laboratory biosecurity, biocontainment, and responsible conduct in the life sciences;

(2) Assist managers, supervisors, and employees to implement applicable statutes and Departmental directives and agency-specific policies; to develop and implement site-specific biorisk management plans, job hazard analyses, and risk assessments; and to provide training and education;

(3) Maintain competence to perform duties described above by attending conferences and training sessions (as funds are available) on biorisk management trends and issues. Some knowledge, skills, and abilities that are minimal requirements may include: performing or assisting with risk assessments and/or job hazard analyses; knowing relevant standards and guidelines; selecting and operating protective equipment; facility design; and disinfection, decontamination, and sterilization methods;

(4) Conduct safety and health site assessments that include biorisk management-related metrics. Assessments should provide guidance on achieving best practices and monitoring compliance with regulatory requirements or agency or location policies and procedures; and

(5) Assist line management to investigate and review incidents involving biological materials to prevent future occurrences.

i. Employees (laboratory, animal facility, field staff) will:

(1) Promote and embody a culture that emphasizes biosafety, biosecurity, and responsible conduct in the life sciences;

(2) Comply with all applicable statutes, policies, and procedures that are specific to the agency, location, or facility;

(3) Read and comply with all safety, health, and biorisk management policies and procedures issued by the location or facility and/or supervisors;

(4) Report all job-related incidents to supervisor, or second line of management when first line supervisor is not available, in a timely manner and in accordance with agency-specific policy. Incidents may include:

(a) Accidents, injuries, illnesses, and exposures (including suspect or potential exposures) to infectious biological materials and biological toxins; and

(b) Security-related incidents and/or environmental releases.

(5) Notify supervisors about the cause or causes of an incident and recommend corrective actions, which may include training;

(6) Promptly report early signs and symptoms of any potentially job-related illnesses and seek medical assistance when necessary;

(7) Properly use and maintain all applicable safety, environmental, and personal protective equipment and clothing;

(8) Participate in all required USDA and agency-specific initiatives and training; and

(9) Make suggestions for improving biosafety, safety, health, and environmental compliance as appropriate.

7. INQUIRIES

All USDA agencies and staff offices shall direct questions and inquiries regarding this DR 4400-007 to the USDA Animal and Plant Health Inspective Service (APHIS) Biosafety Officer at 301-436-3117.

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APPENDIX A

ACRONYMS AND ABBREVIATIONS

ABSA	American Biological Safety Association International
ANSI	American National Standards Institute
APHIS	Animal and Plant Health Inspection Service
ARS	Agriculture Research Service
BMBL	Biosafety in Microbiological and Biomedical Laboratories
BSAT	Biological Select Agent and Toxin
BSL	Biosafety Level
CDC	Centers for Disease Control and Prevention
CFR	Code of Federal Regulations
DNA	Deoxyribonucleic acid
DOT	Department of Transportation
DM	Departmental Manual
DR	Departmental Regulation
DURC	Dual Use Research of Concern
EPA	Environmental Protection Agency
FAO	Food and Agriculture Organization of the United Nations
FS	Forest Service
FSAP	Federal Select Agents Program
FSIS	Food Safety Inspection Service
GAO	Government Accountability Office
HCBA	High Consequence Biological Agents
HVAC	Heating, Ventilation, and Air Conditioning
IATA	International Air Transportation Association
ISO	International Organization for Standardization
JCBMP	Joint Committee on Biorisk Management Policy
JHA	Job Hazard Analysis
LAI	Laboratory-Acquired Infection
NA	Nucleic Acid
NIH	National Institutes of Health
NSF	National Sanitation Foundation
OBC	Operational Biorisk Subcommittee
OCS	Office of the Chief Scientist
OHS	Office of Homeland Security
OIE	World Organization for Animal Health
OSHA	Occupational Safety and Health Administration
OSSP	Office of Safety, Security and Protection
POA&M	Plan of Action and Milestones
POC	Point of Contact
SOP	Standard Operating Procedure
RNA	Ribonucleic acid molecule
PPE	Personal Protective Equipment

UN	United Nations
USDA	United States Department of Agriculture
USPHS	United States Public Health Service
WHO	World Health Organization

APPENDIX B

DEFINITIONS

a. Accident. An unplanned event that results in an injury, illness, or damage to property or the environment involving employees, volunteers, contractors, cooperators, emergency personnel, visitors, or the public.

b. Biocontainment. A term used in describing safe methods, facilities features, and containment or safety equipment for managing infectious materials in the laboratory environment where these materials are being handled or maintained. The purpose of containment is to reduce or eliminate exposure of laboratory workers, other persons, and the outside environment to potential biohazardous agents or toxins.

c. Biohazard. Include certain types of recombinant Deoxyribonucleic Acid (DNA); or synthetic nucleic acid (NA – DNA or Ribonucleic Acid (NA)); biological agents infectious to humans, animals, or plants (e.g., parasites, viruses, bacteria, fungi, and rickettsia); and biologically active agents (i.e., toxins, allergens, prions, and venoms) that may cause disease in other living organisms or significantly affect the environment, community, commerce, or trade agreements.

d. Biohazardous agent. An infectious agent or toxin, or hazardous biological agent, or part thereof, regardless of origin (naturally occurring, bioengineered, or a synthesized component of any such microorganism or infectious substance) that presents a real or potential risk to humans, animals, or plants, either directly through infection, or indirectly through environmental disruption.

e. Biological agent(s). Any microorganism (including, but not limited to, bacteria, viruses, fungi, rickettsiae, or protozoa), or infectious substance, or any naturally occurring, bioengineered, or synthesized component of any such microorganism or infectious substance, capable of causing:

   (1) Death, disease, or other biological malfunction in a human, an animal, a plant, or another living organism;

   (2) Deterioration of food, water, equipment, supplies, or material of any kind; or

   (3) Deleterious alteration of the environment.

f. Biological Select Agent and Toxins. Biological agents and toxins that have the potential to pose a severe threat to public health and safety, animal health and safety, plant health and safety, or to the safety of animal or plant products, regulated under 9 CFR Part 121, 7 CFR Part 331, and 42 CFR Part 73.

g. Biorisk. Effect of uncertainty expressed by the combination of the consequences of an event (including changes in circumstances) and the associated "likelihood" of occurrence, where *biological material* is the source of *harm*.

h. Biorisk management. The effective management or mitigation of risks posed by working with biohazardous agents includes a range of practices and procedures to ensure biosafety, biocontainment, and biosecurity of those biohazardous agents. Biorisk management includes the full spectrum of safety and security measures for laboratories, including hazard identification and assessment, risk communication and training, development of standard operating procedures, and use of personal protective equipment, engineering controls, and facility design.

i. Biorisk management staff. A staff member with an education and professional background in the life sciences and who may hold various titles such as biosafety officer/specialist/manager, biocontainment officer/specialist/manager, or biosecurity officer/specialist/manager. To be effective, biorisk management staff must develop knowledge, skills, and abilities in the principles of epidemiology, disease transmission routes and patterns, biological risk assessment, JHA and risk management strategies, disinfection and sterilization methodology and kinetics, sampling methods and environmental controls (engineering controls; facility design; Heating, Ventilation, and Air Conditioning (HVAC) systems; and laboratory biosecurity). In some instances in which a dedicated biorisk management staff person is not available, a staff member with a background in occupational safety and health or general science disciplines may fill this role as an adjunct or collateral duty provided the staff member has received appropriate training and education.

j. Biosafety. The use of specific practices, safety equipment, and specially designed facilities to ensure that workers, the community, and the environment are protected from infectious agents, toxins, and biological hazards. A biosafety program will identify biological hazards, assess the level of health-related risks of the biological hazards present, and identify ways to reduce the health-related risks associated with the biological hazards.

k. Biosafety levels (BSL). A combination of work practices and physical containment requirements (facility and safety equipment) designed to reduce the risk of a microorganism's ability to cause injury through disease to workers, animals (wild and domestic), the environment, and the public.

l. Biosecurity. Although this term has multiple definitions, for the purpose of this document the term biosecurity will refer to the protection of biological agents or toxins from loss, theft, diversion, or intentional misuse. This is consistent with current USDA Directives, CDC's BMBL, World Health Organization (WHO) Laboratory Biosafety Manual, and American Biological Safety Association (ABSA) International usage of this term.

m. Culture of safety. The common shared and learned meanings, experiences, values, and interpretations of all members of an organization that guide and affect attitudes, patterns of behavior, attention and actions toward improving safety performance, and commitment to risk control on a daily basis.

n. Deoxyribonucleic acid (DNA). A nucleic acid that carries the genetic information in cells and some viruses, consisting of two long chains of nucleotides twisted into a double helix and joined by hydrogen bonds between the complementary bases adenine and thymine or cytosine and guanine. DNA sequences are replicated by the cell prior to cell division and may include genes, intergenic spacers, and regions that bind to regulatory proteins.

o. Facility. Any USDA owned, leased, or shared space where a unit of a USDA agency performs specific services or functions.

p. Field research setting. A site where research is conducted on livestock, free-ranging fish and wildlife populations, habitats and landscapes, watersheds, ecosystems, and other natural resources, and where the field work includes the study and collection of information to include biological sample collection outside a laboratory, library, or interior workplace setting.

q. Federal Select Agent Program (FSAP). A program with shared responsibility by the Animal and Plant Health Inspection Service (APHIS) and the Centers for Disease Control and Prevention (CDC).

r. Field biosafety (outside of laboratory containment). A subset of field safety procedures and protective equipment designed to reduce the risk of occupationally acquired infection, cross-contamination of biological samples, and cross-contamination (through mechanical transfer) to other environmental or field locations when work is conducted outside the controlled laboratory environment and therefore may introduce new infections to naïve animal populations.

s. High Containment Biological Agents and Toxins. Those biological agents and toxins, including transboundary animal diseases and their vectors, which require high or maximum containment facilities.

t. Incident. An occurrence (event), natural or man-made, that happens within the work environment involving biological-associated hazards, including toxins. Incidents can include, but are not limited to, unsafe occurrences and hazardous situations. This could include preventable and non-preventable occurrences; occurrences involving individuals/personnel or situational events not involving individuals/personnel; and may or may not be accidental.

u. Job hazard analysis (JHA). A technique that focuses on job tasks as a way to identify hazards before they occur. It focuses on the relationship between the worker, the task, the tools, and the work environment. The goal is to identify hazards and eliminate or reduce (risk mitigation) them to an acceptable level.

v. Laboratory biosecurity. See Biosecurity definition.

w. Occupationally-acquired infection. An infection that is directly related to the exposure of an individual to an infectious agent as a direct result of performing one's job duties.

(1) Laboratory-acquired infection (LAI). An infection acquired through laboratory or laboratory-related activities regardless of whether it results in symptomatic or asymptomatic conditions.

(2) Field-acquired infection. An infection acquired through field research, diagnostic, or response activities regardless of whether it results in symptomatic or asymptomatic conditions.

x. Pathogen. A bacterium, virus, or other microorganism that can cause disease.

y. Personal protective equipment. Equipment worn to minimize exposure to hazards that cause serious workplace injuries and illnesses. Such injuries and illnesses may result from contact with biological, chemical, radiological, physical, electrical, mechanical, or other workplace hazards. Personal protective equipment may include items such as gloves; safety glasses and shoes; earplugs or muffs; hard hats; respirators; and coveralls, vests, or full body suits.

z. Ribonucleic Acid (RNA). A polymeric molecule implicated in various biological roles in coding, decoding, regulation, and expression of genes. RNA and DNA are nucleic acids, and along with proteins and carbohydrates, constitute the three major macromolecules essential for all known forms of life.

aa. Risk assessment. A process used to identify the hazardous characteristics of a biological agent (known infectious, zoonotic, or environmental hazards), toxins, or material; the activities that can result in a worker's exposure to an agent or material and/or environmental release of a biological agent that would negatively affect agriculture or the environment (whether in a laboratory, vivarium, animal pen, greenhouse, or field activity); the likelihood that such exposure will cause a negative effect such as a laboratory-acquired infection or field-acquired infection; and the probable consequences of such an event. The information identified by the risk assessment will provide a guide for the selection of biosafety levels (laboratory setting or vivarium), appropriate work practices, safety equipment (which includes personal protective equipment), and facility safeguards at which research or diagnostic activities should be conducted. Likewise, a job hazard analysis should be performed to identify and mitigate risks to known or potentially infectious biological agents, toxins, or materials involving field activities where facility safeguards are not feasible. A risk assessment should be performed to provide the information needed to eliminate or reduce a particular risk to an acceptable level. The assessment of risk needs to be carried out before work begins (and repeated when changes are to be made in biohazardous agents or toxins, practices, employees, or facilities/worksite) by a knowledgeable person using professional judgment.

bb. Standard operating procedure (SOP). A set of step-by-step instructions compiled by an organization to help workers carry out routine operations. SOPs aim to achieve efficiency, quality output and uniformity of performance, while reducing miscommunication and failure to comply with industry regulations.

cc. Toxin. A broad range of substances of natural and xenobiotic methods that may cause death or severe incapacitation at relatively low exposure levels. Biological toxins include metabolites of living organisms, degradation products of dead organisms, and materials rendered toxic by the metabolic activity of microorganisms. Examples of toxins include *Staphylococcus* enterotoxins, saxitoxin, cholera toxin, botulinum toxin, ricin, abrin, and mycotoxins.

APPENDIX C

AUTHORITIES AND REFERENCES

7 CFR Part 330.201, *General: Application for Permits to Move Plant Pest*

7 CFR Part 331, *Possession, Use, and Transfer of Select Agents and Toxins* (*Agriculture)*

9 CFR Part 121, *Possession, Use, and Transfer of Select Agents and Toxins* (*Animals and Animal Products)*

9 CFR Part 122, *Organisms and Vectors*

42 CFR Part 71.54, *USPHS Import Regulations for Infectious Biological Agents, Infectious Substances, and Vectors*

42 CFR Part 73, *Select Agents and Toxins (Public Health)*

APHIS, *User's Guide for Introducing Genetically Engineered Plants and Microorganisms*

ARS, ARS Manual 242.1, Chapter 9, *Biohazard Containment Design*, May 2012

CDC, *Biosafety in Microbiological and Biomedical Laboratories*, 5th Edition, December 2009

CRC Press - *Laboratory Biosecurity Handbook*, 2nd Edition, 2007

CWA, Cen Workshop Agreement 15793, *Laboratory Biorisk Management*, 2011

DOT, 49 CFR Parts 171-180, *Transporting Infectious Substances*

IATA, *Dangerous Goods Regulations*, 2019

ISO, *35001 Biorisk Management for laboratories and other related organizations*; 2019

NIH, *NIH Guidelines for Research Involving Recombinant or Synthetic Nucleic Acid Molecules (NIH guidelines)*, April 2019

OIE (World Organization of Animal Health), *Manual of Diagnostic Tests and Vaccines for Terrestrial Animals - International Transfer and Laboratory Containment of Animal Pathogens*, 2018

OSHA, *Occupational, Safety and Health Administration, General Duty Clause*

UN-FAO, *Laboratory Mapping Tool*, 2016

USDA, DM 4400-007, *USDA Biorisk Management Manual*, forthcoming

USDA, DR 9610-001; *USDA Policies and Procedures on Security, Suitability and Incident Response for High Containment Facilities*, forthcoming

USDA DM 9610-001; *USDA Policies and Procedures on Security, Suitability and Incident Response for High Containment Facilities*, forthcoming

USDA, DM 9610-002, *USDA Security Policies and Procedures for Laboratories and Technical Facilities (Excluding Biosafety Level-3 Facilities)*, April 30, 2003, under revision

USDA, DR 9630-001, *USDA Policies and Procedures on Biohazardous Waste Decontamination Management, and Quality Controls at Laboratories and Technical Facilities*, June 18, 2009

USDA, *Joint Committee on Biorisk Management Policy Charter*

USDA, *Joint Committee on Biorisk Management Policy Operational Biorisk Subcommittee Charter*

WHO, World Health Organization *Laboratory Biosafety Manual*, third edition, 2004, under revision